CHEM 515 Practice Session 4: Chemical reactions in solvation environment; excited-state calculation and analysis using TD-DFT

Name: ________________________

In this practice, you will first get some further practice on studying the thermodynamics of chemical reactions with the example of predicting the pKa values of acetic acid and trifluoroacetic acid, for which the use of solvent model will be essential. You will then practice time-dependent density functional theory (TD-DFT) calculations and excited-state analysis with two molecules: (i) pyridine; (ii) dimethylaminobenzonitrile (DMABN).

- For the calculations involved in **Exercises 1 & 3**, you should complete them on the Itcotck server, using the command "`qchem -nt 4 [input] [output] &`". Make sure to put an "`&`" at the end of your command so the job will run in the background.
- To get some practice on running calculations on a supercomputer cluster, you are required to use the SDSC Expanse cluster to complete **Exercise 2**. Using the Itcotck or even the IQmol server, you will still be able to complete this practice, but **1 point will be deducted**.

You will need to **complete the tables and short-answer questions on this handout** and also submit the files as requested. Please refer to the Assignment page on Canvas for detailed submission guidelines. **8 points** are associated with the completion of this handout and the rest **2 points** are for attendance.

Exercise 1: Predicting the pKa values of acetic acid and trifluoroacetic acid

Acetic acid (CH_3COOH) and trifluoroacetic acid (CF_3COOH) are two common organic acids. Their acidity can be quantified using their pKa values, which is defined as the negative logarithm of the equilibrium constant ($K^{\ominus}$) of the dissociation reaction:

$$\mathrm{HA(aq)} \rightarrow \mathrm{H^+(aq)} + \mathrm{A^-(aq)}$$

where "aq" stands for aqueous solution. Based on

$$\Delta G^{\ominus} = -RT \ln K^{\ominus} = -2.303RT \log_{10} K^{\ominus}, \tag{1}$$

we can derive that

$$\mathrm{pKa} = \Delta G^{\ominus}/2.303RT \tag{2}$$

If we convert $\Delta G^{\ominus}$ to be in kcal/mol, then at 298.15 K we have $RT = 0.592$ kcal/mol. Therefore, the problem of pKa prediction is essentially evaluation of the Gibbs free energy change of the acid dissociation reaction in an aqueous solution. Here we employ the **direct method**, in which the electronic energy and thermodynamic corrections will all be calculated under a solvation environment. The theoretical model we employ here is B3LYP/6-31+G(d,p)/C-PCM. For the C-PCM model, the dielectric constant ε will be set to 78.39 for water. Note that the accurate prediction of pKa remains to be a challenging problem in computational chemistry, and it is not surprising at all to get errors of a few pKa units.

We will start with calculating the pKa of CH_3COOH. Note that the Gibbs free energy of proton (H^+) in water is available as a constant: $G(\mathrm{H^+, aq}) = -270.273468$ kcal/mol. Therefore, we only need to calculate the Gibbs free energies of CH_3COOH and its conjugate base CH_3COO^-:

Table 1: Worksheet to determine the pKa values of CH_3COOH and CF_3COOH at the B3LYP/6-31+G(d,p)/C-PCM level of theory

	CH_3COOH(aq)	CH_3COO^-(aq)	CF_3COOH(aq)	CF_3COO^-(aq)
$E_e + E_{solv}$ (a.u.)				
$H^{\ominus}_{vib}$ (kcal/mol)				
$S^{\ominus}_{vib}$ [cal/(mol·K)]				
$G^{\ominus}$ (kcal/mol)				
$\Delta G^{\ominus}$ (kcal/mol)		—		—
pKa		—		—

- For the starting geometry of CH_3COOH, we can load the built-in geometry through "`Add Fragment` → `Molecules` → `Carboxylic Acids` → `Acetic Acid`". Set up a geometry optimization job at the B3LYP/6-31+G(d,p) level (the basis needs to be manually specified), and then apply the C-PCM solvation model by selecting "`Advanced` → `Solvent Model`" and setting the "`Solvent Method`" to "`PCM`". **Note:** you may need to click on "PCM" *twice* to fully apply the model. After doing that, you should see lines

```
$pcm
   THEORY  CPCM
$end

$solvent
   DIELECTRIC  78.39
$end
```

  appear at the bottom of your input. The dielectric constant for water has been loaded by default so you don't need to change any other parameters.

- Now add a second job to perform a vibrational frequency calculation at the optimized geometry (remember to change the `jobtype`). Typically the keywords for the setup of the solvent model in the first job should be carried over to the second by IQmol, but if not, just set up the C-PCM model for water again for the frequency calculation. For both jobs, you should set "`GUI = 0`".

- Now you can paste the generated input to an input file on the server and execute your calculation there. Once successfully finished, by searching for "`Final energy`" in the output as last time you will obtain the $E_e + E_{solv}$ value for CH_3COOH(aq). For the thermodynamic contribution to enthalpy and entropy, you will need to do something different from last time: instead of reporting the "`Total Enthalpy`" and "`Total Entropy`", you should **only include the vibrational contribution here**. This is because the equations for calculating the translational and rotational contributions to enthalpy and entropy were derived for gas-phase molecules, which strictly should not apply here. To find the vibrational contributions, you can search for "`QRRHO-Vib. Enthalpy`" and "`QRRHO-Vib. Entropy`". Here we adopt the values calculated by the quasi-rigid-rotor-harmonic oscillator (QRRHO) model as it has a more proper treatment for vibrational modes below 100 cm^{-1}, which exist in these molecules. Evaluate

$$G^{\ominus} = (E_e + E_{solv}) + H^{\ominus}_{vib} - TS^{\ominus}_{vib} \tag{3}$$

  at $T = 298.15$ K and report the values in **Table 1**.

- To set up the calculation for CH_3COO^-, you can simply make a copy of the input file for acetic acid using the "`cp`" command on the server (remember that the input file for acetate should be renamed). Then you just need to delete the protonic H from the input coordinates and set the charge and multiplicity to "−1 1".

Report the computational results for CH_3COOH and CH_3COO^- in Table 1 and calculate the pKa value of acetic acid using Eq. (2). (0.75 pt)

For CF_3COOH, you should start from the provided "`TfAcOH.xyz`" that was downloaded from the CC-CBDB database rather than the built-in trifluoroacetic acid structure in IQmol since the latter turns out to be an energetically less favorable rotamer. The rest of the procedure should be pretty much *the same* as for the acetic acid. **Complete the last two columns of Table 1 (0.75 pt), and then answer the following questions:**

- The experimental pKa values for acetic and trifluoroacetic acetic acids are **4.76** and **0.23**, respectively. Did your calculations correctly predict the qualitative trend of the acidity of these two acids? **(0.25 pt)** Quantitatively, how many pKa units do your computational results differ from the experimental values? Report the errors (with sign) for both acids. **(0.25 pt)**

- One possible source of error is the employed electronic structure method. Just like what we did for the reaction barrier of the Diels-Alder reaction in our last practice, here we are going to refine the $E_\mathrm{e} + E_\mathrm{solv}$ term at the ωB97M-V/def2-TZVPPD/C-PCM level of theory, i.e., using a more accurate density functional and a larger basis set. Using **Table 2** below as a worksheet, calculate the pKa values using the refined $E_\mathrm{e} + E_\mathrm{solv}$ values, and then discuss whether this has improved the accuracy of pKa predictions or not. **(0.5 pt)**

- **(0.5 bonus point)** Repeat the energy refinement calculations using the same functional and basis set (ωB97M-V/def2-TZVPPD) but with the SMD solvation model. Do your calculations on Excel and report the pKa values you obtained below. **Then answer the following questions**:
  (i) What additional physical component does SMD incorporate while C-PCM doesn't? (ii) Does SMD provide better pKa predictions for these two acids?

  Note: To set up the SMD model, go to "`Advanced` → `Solvent Method`" and select "`SMD`" under "Solvent Method": you may also need to click on it twice to fully load it. You should see these lines appear at the bottom of your input:

  ```
  $smx
     SOLVENT  water
  $end
  ```

Table 2: Worksheet to determine the pKa values of CH_3COOH and CF_3COOH with the refinement at the ωB97M-V/def2-TZVPPD/C-PCM level

	CH_3COOH(aq)	CH_3COO^-(aq)	CF_3COOH(aq)	CF_3COO^-(aq)
$E_e + E_{solv}$ (a.u.)				
$G^\ominus$ (kcal/mol)				
$\Delta G^\ominus$ (kcal/mol)		—		—
pKa		—		—

Exercise 2: Singlet and triplet excited states of pyridine via TD-DFT calculations

In this practice, we will use time-dependent density functional theory (TD-DFT) to investigate the singlet and triplet excited states of pyridine. Note that the Tamm-Dancoff approximation (TDA) will be used throughout this exercise, which is the default option in Q-Chem. All the excited state calculations should be done based on the given structure "`pyridine.xyz`", i.e., no geometry optimization should be done in this exercise. **Note:** You are required to use **SDSC Expanse** to complete the calculations in this exercise.

We will start with calculating **8 singlet states** using ωB97X-D/aug-cc-pVDZ. Using IQmol, the job can be set up as follows:

- Open up the XYZ file and go straight to "`Q-Chem Setup`". Under `Method`, find and select "`TD-DFT`". Once you did that, some additional job control options should appear under "`CIS/TD-DFT`" at the left bottom of the control panel. Now you should select (or type) "`omegaB97X-D`" under "`Exchange`" and change the basis set to aug-cc-pVDZ. (**Note:** "`Exchange = omegaB97X-D`" + "`Correlation = None`" is the old way to invoke the ωB97X-D functional in Q-Chem inputs, which has been deprecated but still works; alternatively, you can simply set "`Method = wB97X-D`".)

- Under the "`CIS/TD-DFT`" options, you should **unselect** "`Triplets`" and set "`Number of Roots`" to 8 so it will calculate 8 singlet states. We will need to use natural transition orbitals (NTOs) to identify the characters of excited states later, so you need to set "`NTO PAIRS`" to 2. Lastly, set "`Convergence`" to 8 under "`SCF control`" and keep "`GUI = 2`". The generated input file should look as below:

```
$molecule
0 1
  C    0.0000000    0.0000000   -1.4099987
  C    0.0000000    1.1932681   -0.6988838
  C    0.0000000   -1.1932681   -0.6988838
  C    0.0000000    1.1398403    0.6914739
  C    0.0000000   -1.1398403    0.6914739
  N    0.0000000    0.0000000    1.3905662
  H    0.0000000    0.0000000   -2.4905277
  H    0.0000000    2.1472345   -1.2045420
  H    0.0000000   -2.1472345   -1.2045420
  H    0.0000000    2.0535202    1.2718329
  H    0.0000000   -2.0535202    1.2718329
$end

$rem
   BASIS  =  aug-cc-pVDZ
   CIS_N_ROOTS  =  8
   CIS_TRIPLETS  =  0
   EXCHANGE  =  omegaB97X-D
   GUI  =  2
   NTO_PAIRS  =  2
   SCF_CONVERGENCE  =  8
$end
```

- Copy the generated text to an input file on the SDSC Expanse cluster, then submit this job using
  `submit_qchem -p 16 -q debug pyridine_singlet_wb97xd_adz.in`
  This means running the calculation with 16-thread parallelization on the "`debug`" queue, which is suitable for quick jobs that can finish within 30 minutes.

Once the TDDFT/TDA calculation finishes, you should copy the input, output, and the checkpoint (`.fchk`) files back to your local computer, and then open the output and checkpoint files using IQmol. The text output contains the information about each excited state, most of which can also be viewed by double-clicking "`Excited States`" under IQmol's "Model View", while through the checkpoint file one can identify the character of each excited state by visualizing the **dominant NTOs** (also known as HONTO and LUNTO), which can be found under "`Surfaces`". Here based on the shapes of NTOs, the excited states can be divided into three categories: (i) $n \rightarrow \pi^*$, for which the HONTO features significant participation of N's lone pair; (ii) $\pi \rightarrow \pi^*$, for which the HONTO and LUNTO are both localized on the π-system; and (iii) Rydberg states, which feature spatially very diffuse LUNTOs.

Based on the analysis of the eight excited states, **Collect the information of the $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ excited states in Table 3 (1 pt)**. **Note:** you should NOT include any Rydberg states in **Table 3**. If the table has too many rows, just leave the extra ones blank; if you need more rows, add more at the bottom.

Table 3: Information about the singlet valence excitations of pyridine calculated using ωB97X-D/aug-cc-pVDZ. Keep **4 digits** after decimal point for both excitation energies and oscillator strengths. Under "State character" you should write "$n \rightarrow \pi^*$" or "$\pi \rightarrow \pi^*$, and under "Dominant Orbitals" you should write something like "H→L", "H−1→L+1", etc.

State index	Excitation energy (eV)	Oscillator strength	State character	Dominant Orbitals

Complete the following:

1. When comparing the $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ states, which type of state is lower in energy? Which type is brighter (i.e., of larger oscillator strength)? **(0.25 pt)**

2. Generate plots of the dominant NTO pairs (HONTO and LUNTO) for the $n \rightarrow \pi^*$ states. **Submit these images with your report**. **(0.25 pt)** (**Note:** there should be a pair of orbitals plotted separately for each excited state.)

3. **Generate an absorption spectrum (in nm) using IQmol**. To do that, first make sure that your IQmol version is 3.1.2 or above (the older versions can't plot spectrum in nm). Double-click on "`Excited States`" and change the unit for the spectrum to nm. Make sure that the spectral lines are roughly centered on the spectrum (you can adjust it by moving left/right or zooming in/out on the plotting area). Finally, set the broadening method to "`Gaussian`". **Save the broadened spectrum and submit that with your report (0.25 pt)**.

Let's now set up another Q-Chem job to calculate **6 triplet states**. To do that, you can make a copy of the input you just used for singlet excited states and then change the line "`CIS_TRIPLETS = 0`" to "`CIS_SINGLETS = 0`", and change the value of `CIS_N_ROOTS` to `6` (of course, you can also use IQmol to set this up from scratch). Perform this calculation using two different density functionals: (i) ωB97X-D; (ii) B3LYP. For each obtained excited state (all should be valence excitations), identify their characters ($n \to \pi^*$ or $\pi \to \pi^*$) by visualizing the NTOs and then report the excitation energies based on the order in **Table 4**: **two $n \to \pi^*$ states go first, followed by four $\pi \to \pi^*$ states. (1 pt)**

Table 4: Calculation of 6 triplet excited states of pyridine (ω stands for excitation energy)

State character	E_{ex}(ωB97X-D) (eV)	E_{ex}(B3LYP) (eV)	E_{ex}(ref) (eV)
$n \to \pi^*(1)$			4.46
$n \to \pi^*(2)$			5.36
$\pi \to \pi^*(1)$			4.30
$\pi \to \pi^*(2)$			4.79
$\pi \to \pi^*(3)$			5.04
$\pi \to \pi^*(4)$			6.24
RMSE			—

Complete the following:

1. Based on the reference values, calculate the root-mean-square errors (RMSE $= \sqrt{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2/N}$) of ωB97X-D and B3LYP **for these 6 triplet states** using Excel and report the values in **Table 4**. **(0.25 pt)** Based on the RMSEs, which functional is more accurate for pyridine's first 6 triplet excited states? **(0.3 pt)**

2. Compared to the reference data, discuss whether the two functionals correctly predict the character of the lowest-energy triplet state. **(0.2 pt)**

3. Comparing the singlet and triplet excitation energies for the two $n \to \pi^*$ states, are the triplet states of higher or lower energies? Does that agree with your expectation? **(0.25 pt)**

Exercise 3: Intramolecular charge-transfer states for twisted DMABN

Dimethylaminobenzonitrile (DMABN) is a commonly used model system for studying *twisted intramolecular charge transfer* (TICT) states. In this practice, you will use three different functionals to calculate the **two lowest singlet excited states** of *twisted* DMABN (of C_{2v} symmetry). You will also practice using Q-Chem's `libwfa` module to perform wavefunction analysis and visualization of excited states.

Since you have gained some experience in setting up TDDFT calculations from **Exercise 2**, I will provide less detailed instructions here. First, with the provides XYZ file "`twisted_DMABN.xyz`", use B3LYP/aug-cc-pVDZ to calculate **4 singlet excited states** and report the excitation energies for the first two states in Table 5. **(0.25 pt)** Make sure that you have `NTO_PAIRS = 2` and `GUI = 2` on in your input so that you can visualize the excited states you obtain. You are also recommended to set "`METHOD = B3LYP`" instead of "`Exchange = B3LYP`" for this practice so that the input can be easily modified to run the other two functionals.

Note: you can also run this job on SDSC Expanse, where the turn around will be faster. However, since the version of Q-Chem on Expanse is slightly older (6.0.2), you will need to add a keyword `symmetry = false` to your input file to turn off integral symmetry (turned off by default in newer versions of Q-Chem). Otherwise, you may encounter some unpleasant SCF convergence issues.

Based on visualization of the NTOs you just obtained, use your words to briefly describe the **charge transfer character** associated with the S_1 and S_2 states (just imagine what you would write if you are discussing these excited states in a paper, e.g., "*electrons are transferred from moiety X to functional group Y*"). **(0.25 pt)**

From the KS-DFT lectures, you should be aware that B3LYP is a global hybrid functional with 20% Hartree-Fock exchange. Now perform the same calculation with BLYP (a semi-local GGA) and CAM-B3LYP (a range-separated hybrid) and report the S_1 and S_2 excitation energies in the same table. **(0.25 pt)** You just need to create a copy of the input you just run, and change the "`METHOD`" to "`BLYP`", and then "`CAM-B3LYP`".

Table 5: The excitation energies (in eV) of the S_1 and S_2 states of DMABN

	BLYP	B3LYP	CAM-B3LYP	Reference
S_1				4.11
S_2				4.74

Do these two functionals yield better or worse agreement with the reference values compared to B3LYP? Rationalize the trend you observe based on the discussion of TDDFT for charge-transfer excited states in class. **(0.5 pt)**

To wrap up this exercise we will perform some excited-state wavefunction analyses based on the CAM-B3LYP results. You can make a copy of the input you have for the CAM-B3LYP calculation and then do the following:

- Delete `NTO_PAIRS = 2` and `GUI = 2`
- Add the following 4 keywords to the `$rem` section:

- STATE_ANALYSIS = TRUE
- MAKE_CUBE_FILES = TRUE
- PLOTS = TRUE
- CIS_RELAXED_DENSITY = TRUE

- Add the following section to the bottom of your input to set up a plotting grid:

```
$plots
    grid_points 80 80 80
$end
```

You can then run the calculation on the server. Once the job is done, a folder "`[jobname].plots`" will be generated along with the output file, under which you can find a big number of `.cube` files organized as `S_1`, `S_2`, ...

Report the dipole moments (in Debye) for the S_1 and S_2 states calculated using their *relaxed* densities in Table 6. To find that, first search for the line "`Analysis of Relaxed Density Matrices`" in your output; then under "`Singlet 1 (relaxed)`" and "`Singlet 2 (relaxed)`", find "`Dipole moment [D]:`". The number after corresponds the total magnitude of dipole moment which you should report. In the same table also **report the ground state dipole** (search for "`Ground State (Reference)`" and scroll down). **(0.25 pt)**
Note: the difference dipole between the ground and excited states for a charge-transfer excitation is usually quite substantial.

Table 6: Dipole moments for the ground and S_1 and S_2 excited states of DMABN

	Ground State	S_1	S_2
Dipole (D)			

Finally, visualize the attachment and detachment densities for the S_1 and S_2 states and **submit the plots with your report (0.25 pt)**. Make sure that

1. The files you visualize are `S_1_rlx_detach.cube`, `S_1_rlx_attach.cube`, `S_2_rlx_detach.cube`, and `S_2_rlx_attach.cube`
2. Use a 10× smaller isosurface value (0.002) to generate these plots
3. The detachment densities are in blue (ideally somewhat transparent) and attachment densities in red so that they can be distinguished. You can use the "`Swap Colors`" feature (double-click "`Cube Data`" under Model View) in IQmol to change the colors if IQmol doesn't give you the correct color in the first place.
4. After you are done with plotting (or have copied everything to your local computer), **delete** the "`[jobname].plots`" folder on the server using the "`rm -r [jobname].plots`" command because those cube files take a lot of disk space.